KRIOS INFRASTRUCTURE

Technical Home Assignment for Geospatial Expert

This task is designed to take 3-4 hours. We are not looking for a polished production system, rather looking for how you think, how you structure your work, and how you communicate trade-offs. A clean, well-reasoned partial answer is more valuable than an overengineered one.

Document your assumptions, what you would do differently with more time, and any known limitations in your outputs. That commentary is as important to us as the code or maps.

Dataset stack

You will work with the following eight layers. These reflect the real data sources KRIOS uses or plans to use. Part of the task is knowing where to find them and how to handle their quirks with given suggestions.

Layer	Source	Comments
Grid capacity & headroom	Fingrid	use Sähkön kulutuskapasiteetti layer
Electricity network	OSM	High-voltage lines and substations
Generation mapping	OSM & other	Power plant locations and type
Data center mapping	OSM, DCD, BroadGroup or other	Existing DC locations for competitive context and interest
Urban centers (100k+)	Statistics Finland or other	Proximity signal for workforce
Land parcels (property boundaries) ≥10 ha	MML (National land survey of Finland) or open topographic data	Core candidate site layer
Gradient (<8%)	MML (National land survey of Finland) or open topographic data	Constraint for cost-effective development
Exclusion zones	Natura 2000 (EEA), Finnish flood hazard maps (SYKE — Finnish Environment Institute), and any other	Fatal flaw

Task

Task reflects real work you would own from day one and your goal is to demonstrate that you can turn raw geospatial data into a ranked, defensible site shortlist that a development team can act on.

Choose a region in Finland as your area of interest (AOI). We suggest starting with a 50–100km radius around a major urban center (Helsinki, Tampere, Oulu, or Turku) and proximity to 220 kV and 400 kV network, but you are free to choose differently — just explain why.

Part 1 — Data Ingestion & Pipeline

Ingest and prepare majority of the layers listed above for your AOI. We expect to see:

- A reproducible Python script that fetches or reads your source data (hardcoded file paths are fine; a working pipeline is better)
- CRS harmonization — all layers in the same projection before any spatial operations
- A reproducible ETL step showing how raw source data is cleaned, transformed, and made ready for spatial analysis
- Basic data quality notes — what is missing, outdated, or unreliable, and how you handled it
- A brief note on what you would automate or improve in a production version

Part 2 — Site Suitability Scoring

Using your prepared layers, implement a two-stage suitability model:

Stage 1 — Fatal flaw filter (hard exclusions)

- Remove all parcels inside Natura 2000 boundaries
- Remove all parcels inside flood hazard zones
- Remove all parcels under 10 ha
- Any additional hard constraints you identify and justify

Stage 2 — Opportunity scoring (weighted criteria)

Score remaining parcels across at least three of the following dimensions. Define your own weights and document your reasoning:

- Grid capacity headroom (MW available at nearest substation with headroom)
- Distance to 220/400 kV+ network connection point
- Distance to nearest 100k+ urban center
- Distance to existing data centers (positive or negative signal — your call)
- Proximity to generation assets (your reasoning required)
- Parcel size above 10 ha threshold (larger = higher score, no larger than 100 ha)
- Optional - land zoning type (preference for industrial)

Produce a ranked output: a GeoJSON or table of your top candidate sites with scores, weights, and the key attributes that drove the ranking.

Part 3 — Map Output & Communication

Produce a map that communicates your results clearly to a non-technical audience — a development director or investor who needs to understand which sites are worth pursuing and why.

Your map should show:

- Your AOI and the key infrastructure layers (grid network, substations with capacity headroom highlighted)
- Exclusion zones (Natura 2000, flood) clearly marked
- Existing DC locations for competitive context
- Top-ranked candidate sites, scored and labelled

- A legend and a short caveats note

Use any tool you are comfortable with: Leaflet, Folium, Kepler.gl, QGIS, Mapbox, or a static export. Include a short note on how you would make this self-serve for the wider team without GIS desktop tools.

Bonus (Optional — Only If Time Allows)

If you finish early, any of the following would strengthen your submission:

- Automate a refresh check for the Fingrid capacity layer — how would you detect when it has been updated and trigger a re-run?
- Sketch a schema design: how would you store these seven layers in PostGIS for multi-country use with versioning and lineage?

Submission Format

Choose whichever format best reflects how you work:

- **Option A — GitHub Repo**: A repo with your scripts, a README explaining how to run them, output files (GeoJSON, scored table, map screenshot or HTML), and a DECISIONS.md noting your assumptions, weights, and trade-offs.
- **Option B — Written Deliverable:** A PDF or Word document with your methodology, annotated map outputs, scored site table, and inline code snippets. DECISIONS commentary must still be present.

Good luck — we look forward to seeing your work and discussing it with you.

Send your submission to manohar.bayyapureddy@kriosinfra.com by the noon on 28.04.2026 (Tue)